

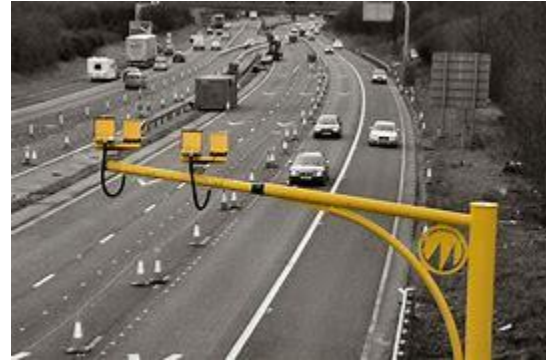
Machine Learning



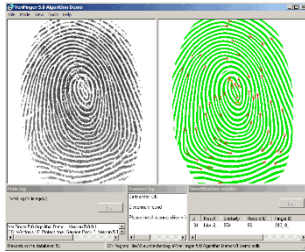
Images to Text

- Reading license plates, zip codes, checks

3 6 8 1 7 9 6 6 9 1
6 7 5 7 8 6 3 4 8 5
2 1 7 9 7 1 2 8 4 5
4 8 1 9 0 1 8 8 9 4
7 6 1 8 6 4 1 5 6 0
7 5 9 2 6 5 8 1 9 7
2 2 2 2 2 3 4 4 8 0
0 2 3 8 0 7 3 8 5 7
0 1 4 6 4 6 0 2 4 3
7 1 2 8 1 6 9 8 6 1



Biometrics



Fingerprint scanners on many new laptops, other devices



Face recognition systems now beginning to appear more widely
<http://www.sensiblevision.com/>

Face detection



- Many new digital cameras now detect faces
 - Canon, Sony, Fuji, ...

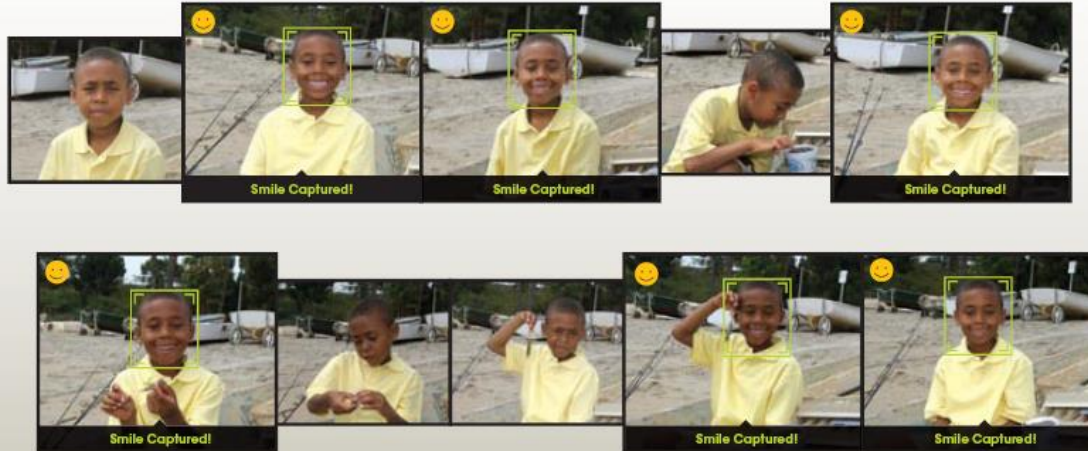
Face recognition: Apple iPhoto, Facebook, Google, etc



Smile detection

The Smile Shutter flow

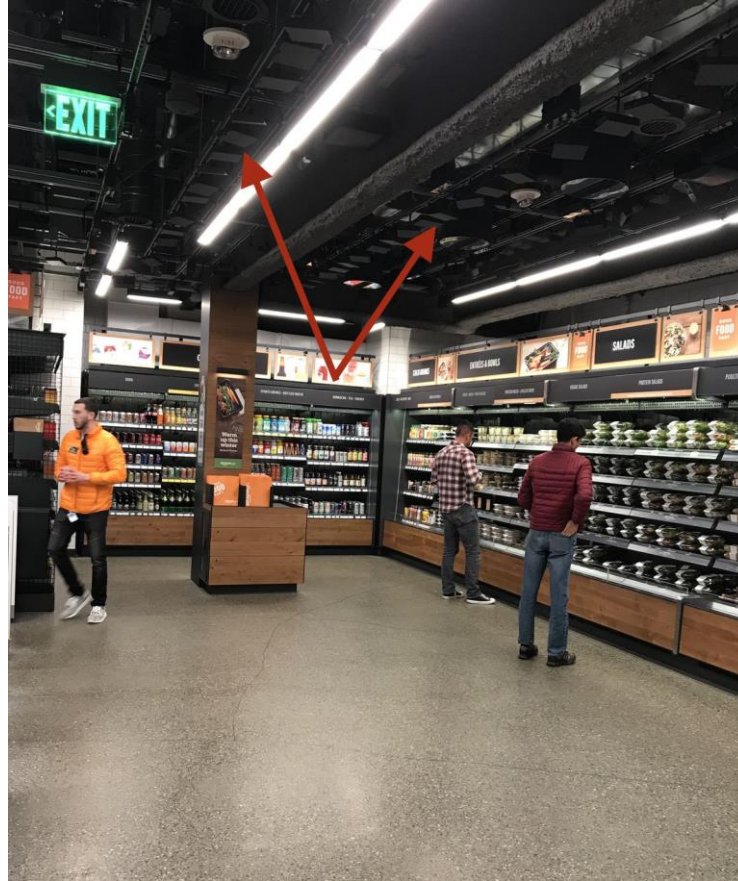
Imagine a camera smart enough to catch every smile! In Smile Shutter Mode, your Cyber-shot® camera can automatically trip the shutter at just the right instant to catch the perfect expression.



Amazon Go



Amazon Go



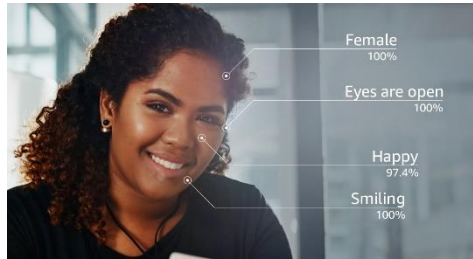
Amazon Rekognition (AWS)



Object
Detection



Facial
Recognition



Facial
Analysis



Pathing



Celebrity
Recognition



Unsafe Content
Detection



Text in
images

Google cars

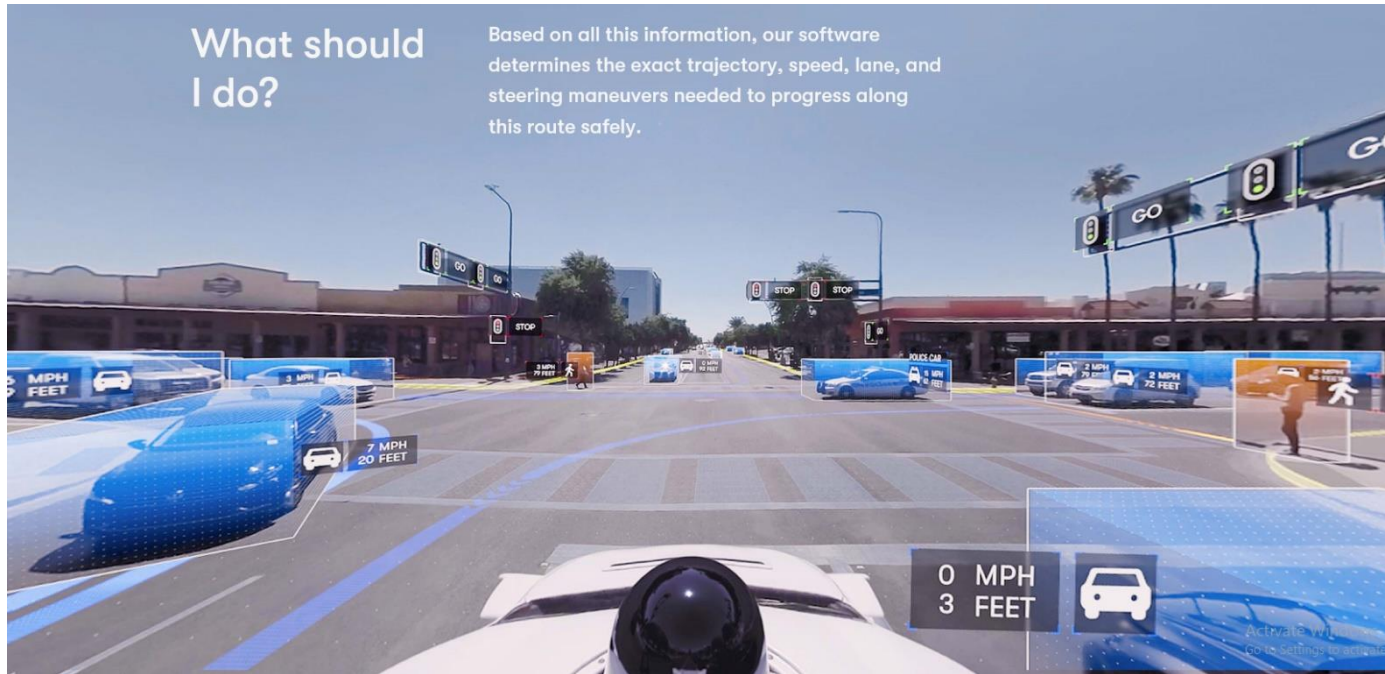


Oct 9, 2010. ["Google Cars Drive Themselves, in Traffic"](#). [The New York Times](#). John Markoff
June 24, 2011. ["Nevada state law paves the way for driverless cars"](#). [Financial Post](#).
Christine Dobby
Aug 9, 2011, ["Human error blamed after Google's driverless car sparks five-vehicle crash"](#). [The Star](#) (Toronto)

WAYMO LLC



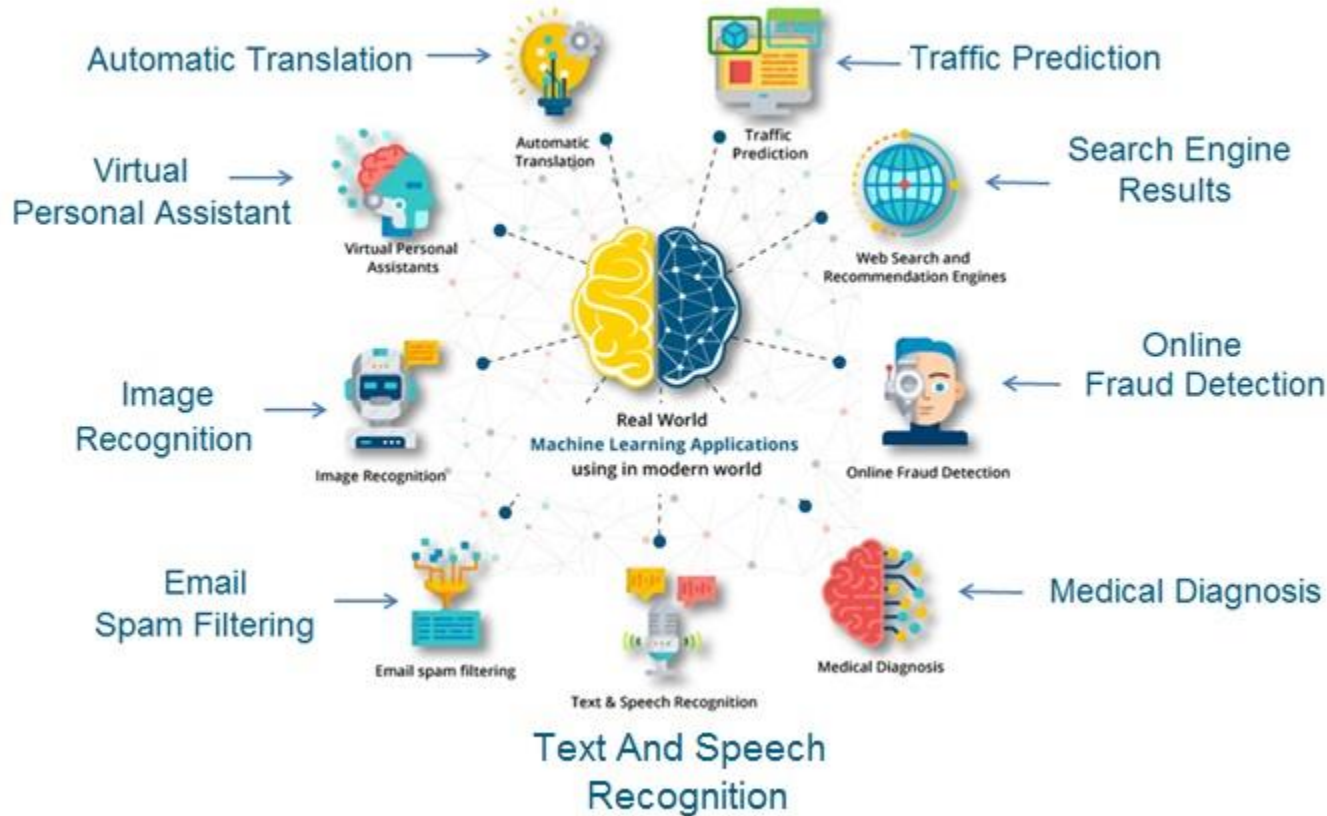
WAYMO Self Driving Car



Tesla Auto Pilot



Real World Applications Of Machine Learning



Machine Learning

- Grew out of work in AI
- New capability for computers

Examples:

- Database mining
 - Large datasets from growth of automation/web.
 - E.g., Web click data, medical records, biology, engineering
- Applications can't program by hand.
 - E.g., Handwriting recognition, most of Natural Language Processing (NLP), Computer Vision.

Machine Learning

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- New capability for computers

Examples:

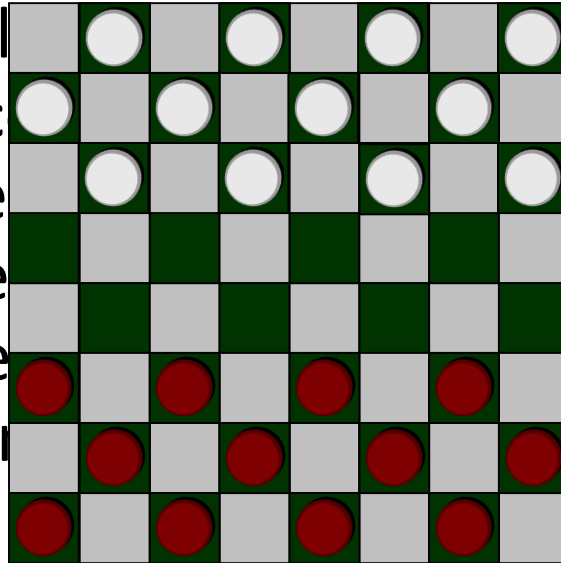
- Database mining
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 - E.g., Web click data, medical records, biology, engineering
- Applications can't program by hand.
 - E.g., Handwriting recognition, most of Natural Language Processing (NLP), Computer Vision.
- Self-customizing programs
 - E.g., Amazon, Netflix product recommendations
- Understanding human learning (brain, real AI).

What is machine learning?

Machine Learning definition

- Arthur Samuel (1959). Machine Learning: Field of study that gives computers the ability to learn without being explicitly programmed.

- Tom Mitchell
Problem: A computer program is said to *learn* from experience E to solve some task T if its performance P , as measured by P , improves with experience.



“A computer program is said to *learn* from experience E with respect to some task T and some performance measure P , if its performance on T , as measured by P , improves with experience E .”

Suppose your email program watches which emails you do or do not mark as spam, and based on that learns how to better filter spam. What is the task T in this setting?

Classifying emails as spam or not spam.

Watching you label emails as spam or not spam.

The number (or fraction) of emails correctly classified as spam/not spam.

None of the above—this is not a machine learning problem.

Machine learning algorithms:

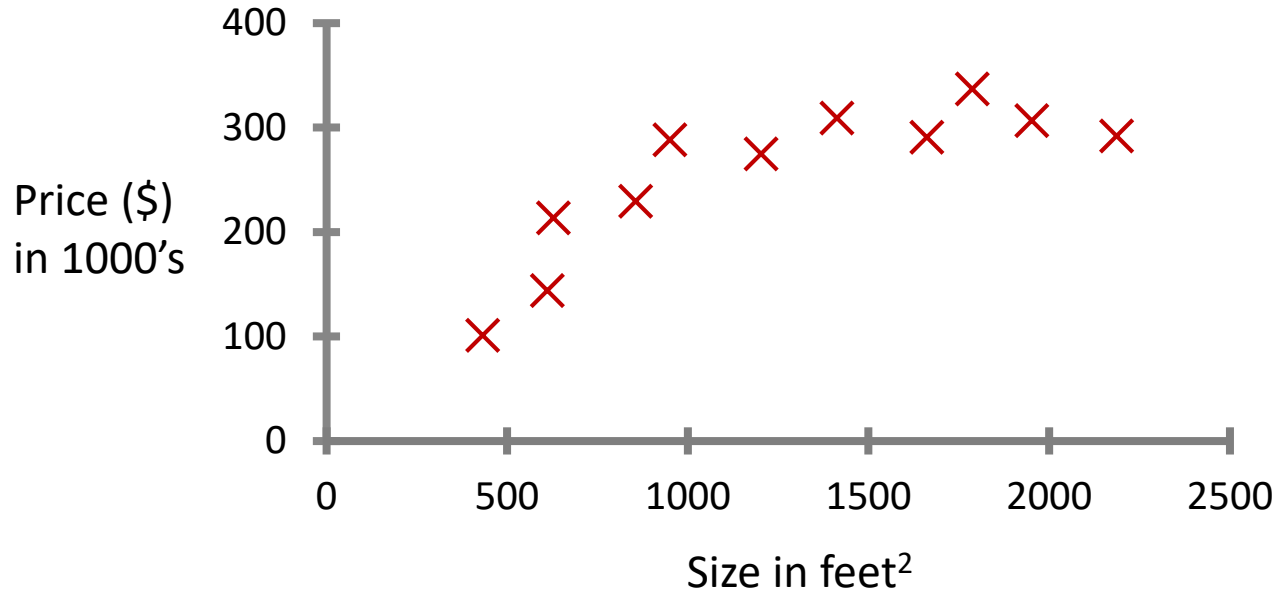
- Supervised learning
- Unsupervised learning

Others: Reinforcement learning, recommender systems.

Also talk about: Practical advice for applying learning algorithms.

Supervised Learning

Housing price prediction.

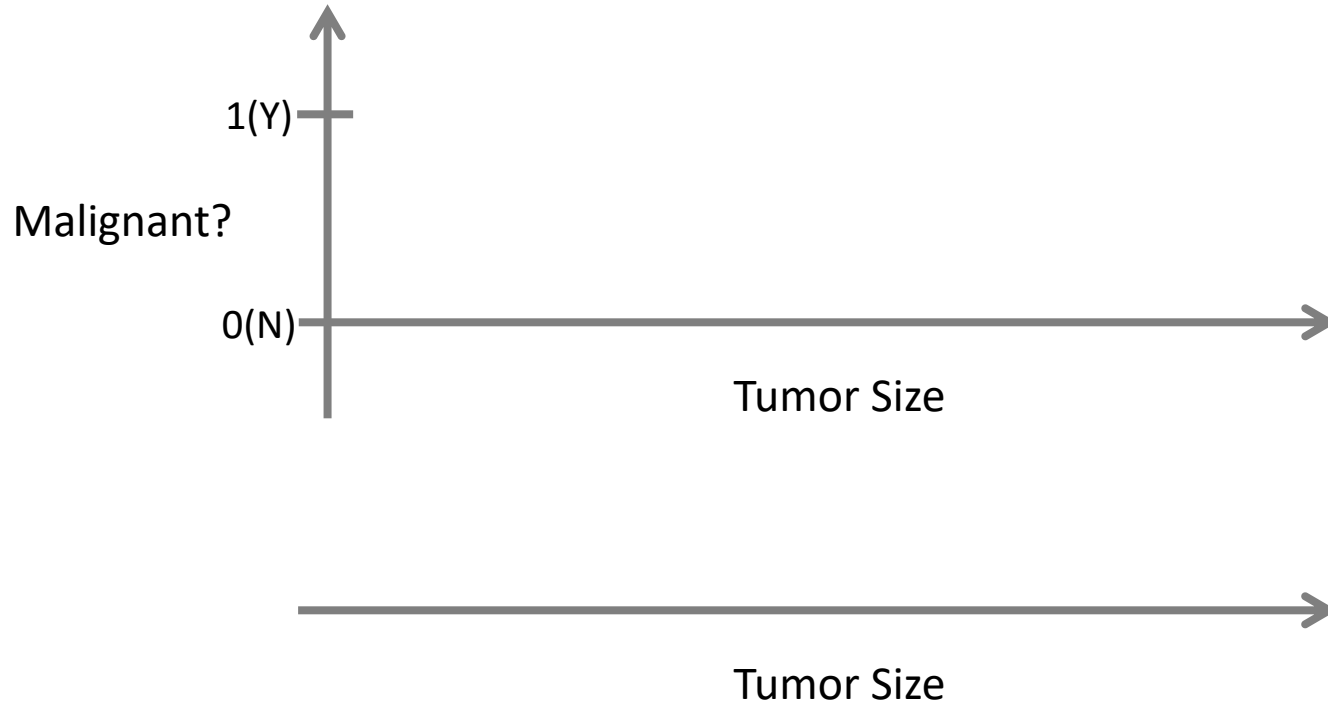


Supervised Learning

“right answers” given

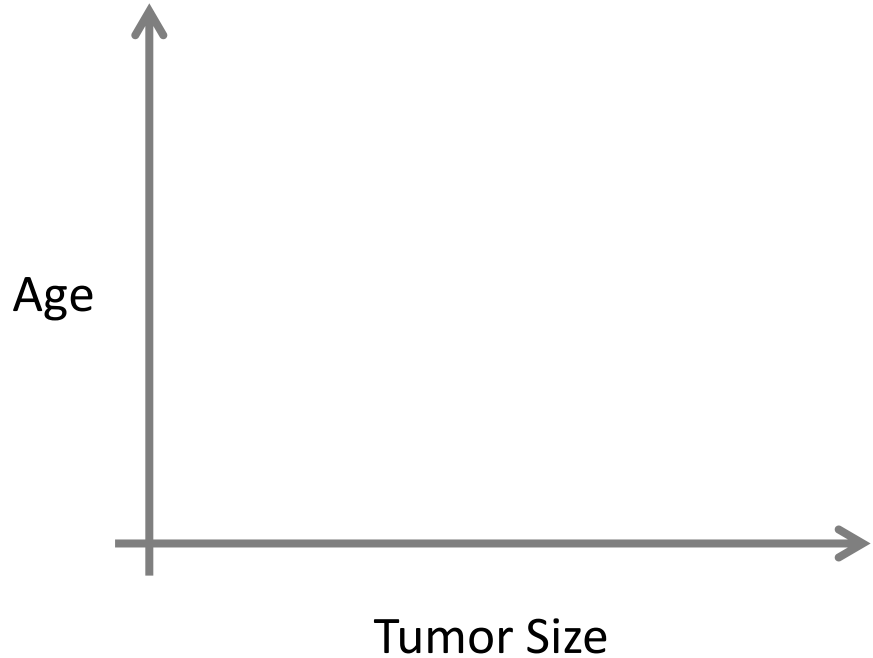
Regression: Predict continuous
valued output (price)

Breast cancer (malignant, benign)



Classification

Discrete valued
output (0 or 1)



- Clump Thickness
- Uniformity of Cell Size
- Uniformity of Cell Shape
- ...

You're running a company, and you want to develop learning algorithms to address each of two problems.

Problem 1: You have a large inventory of identical items. You want to predict how many of these items will sell over the next 3 months.

Problem 2: You'd like software to examine individual customer accounts, and for each account decide if it has been hacked/compromised.

Should you treat these as classification or as regression problems?

Treat both as classification problems.

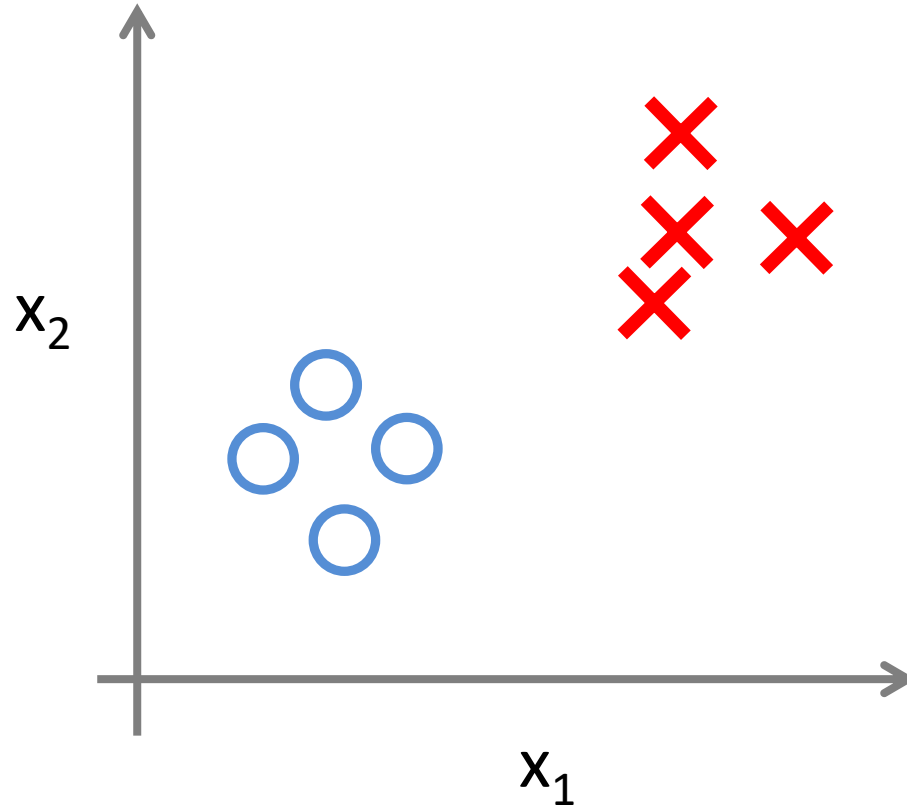
Treat problem 1 as a classification problem, problem 2 as a regression problem.

Treat problem 1 as a regression problem, problem 2 as a classification problem.

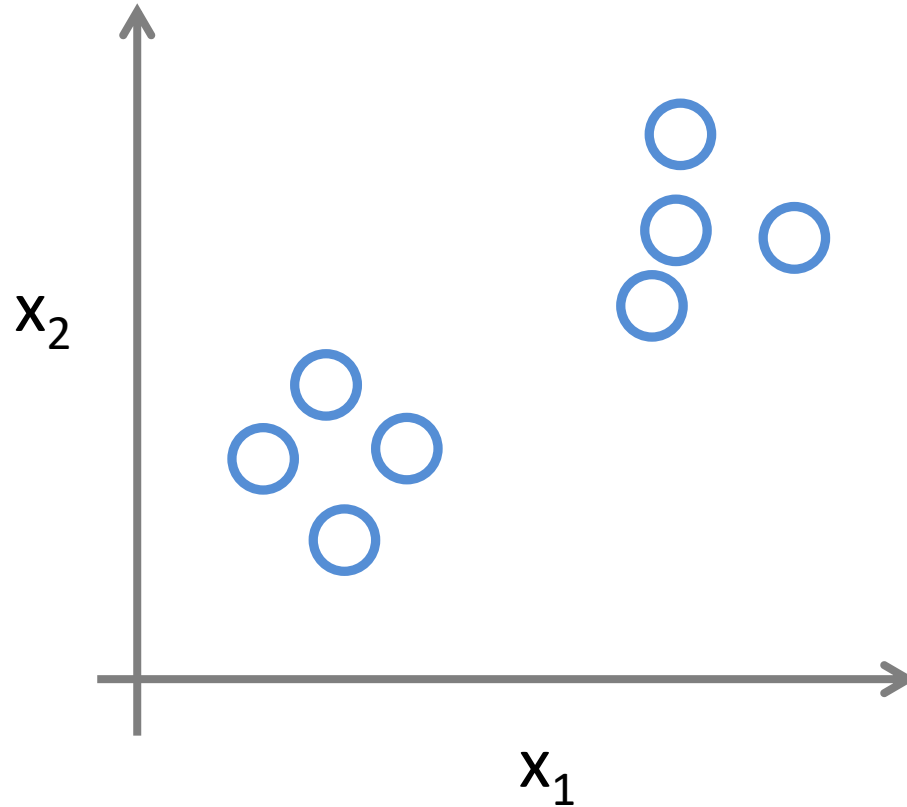
Treat both as regression problems.

Unsupervised Learning

Supervised Learning



Unsupervised Learning



BP Kills Macondo, But Its ...

blogs.wsj.com/source/2010/09/20/bp-kills-macondo-but-its-legacy-lives-on/

THE WALL STREET JOURNAL.
Digital Network

WSJ.com

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SEPTEMBER 20, 2010, 12:44 PM GMT

BP Kills Macondo, But Its Legacy Lives On

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- Will Banks Be Forced to Split Retail And Banking Arms?
- Timing of Ratings Award Intriguing
- BP Kills Macondo, But Its Legacy Lives On
- We Already Need a Sequel to Basel III

By James Herron

BP confirmed late Sunday that the Macondo well that leaked almost five million barrels of oil into the Gulf of Mexico has been permanently sealed, but the well will continue to affect BP and the wider oil industry for many years.

The most immediate worry for BP and its shareholders is how the authorities will apportion blame for the spill. BP's own investigation spread responsibility across

Associated Press

Fire boat response crews battled the blazing remnants of the off shore oil rig Deepwater Horizon on April 21, 2010.

Allen: Well is dead, but m... x

edition.cnn.com/2010/US/09/20/gulf.oil.disaster/

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Allen: Well is dead, but much Gulf Coast work remains

By the CNN Wire Staff
September 20, 2010 -- Updated 1317 GMT (2117 HKT)

Click to play

What next for Gulf oil spill?

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More

STORY HIGHLIGHTS

(CNN) -- The ruptured Macondo well, a mile under the Gulf of Mexico off the Louisiana coast, has been pronounced dead.

BP oil spill cost hits nearl...

[www.guardian.co.uk/environment/2010/sep/20/bp-oil-spill-deepwater-horizon-costs-10bn](#)

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BP oil spill cost hits nearly \$10bn

BP has set up a \$20bn compensation fund after the Deepwater Horizon disaster, which has so far paid out 19,000 claims totalling more than \$240m

Julia Kollewe

guardian.co.uk, Monday 20 September 2010 08.33 BST

Article history



BP's costs for the Deepwater Horizon disaster have hit \$10bn. Photograph: Ho/Reuters

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US government declares well 'dead'

BP oil spill news on Twitter

Latest news on the BP oil spill in the Gulf of Mexico

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BP_America: Find fact sheets, open house info and e-mail alert sign-ups at @EPAgov's oil spill community page: <http://bit.ly/diuHoH>

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guardianeco: UN report on Nigeria oil spills relies too much on data from Shell | Nnimmo Bassey <http://bit.ly/dBd7Ru>

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BP_America: Newly discovered microbe thriving from consumption of #oil in the #Gulf of Mexico: <http://bit.ly/9kQYwa>

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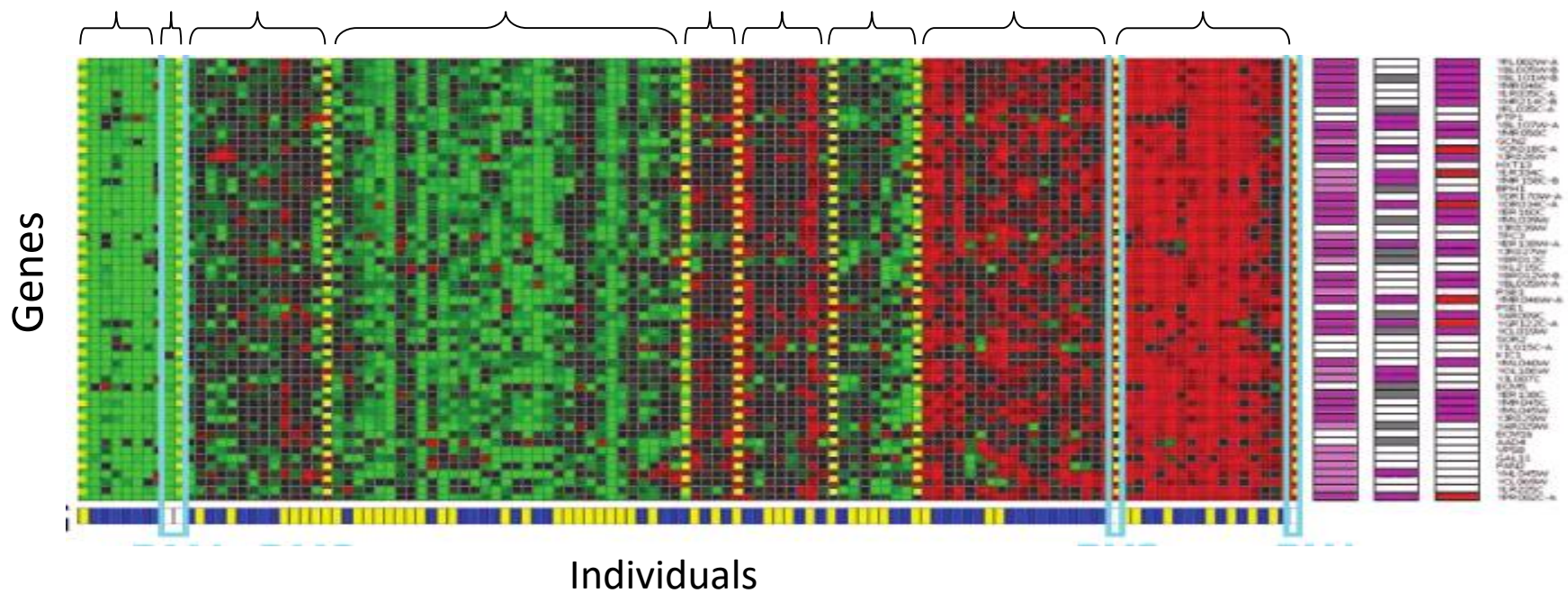
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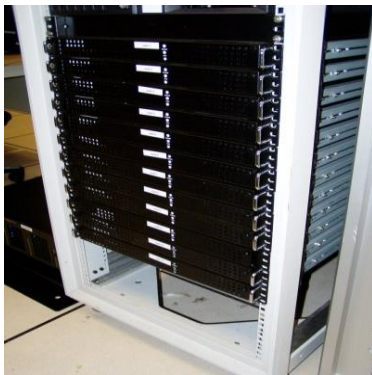
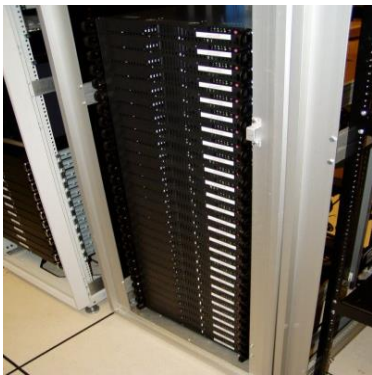
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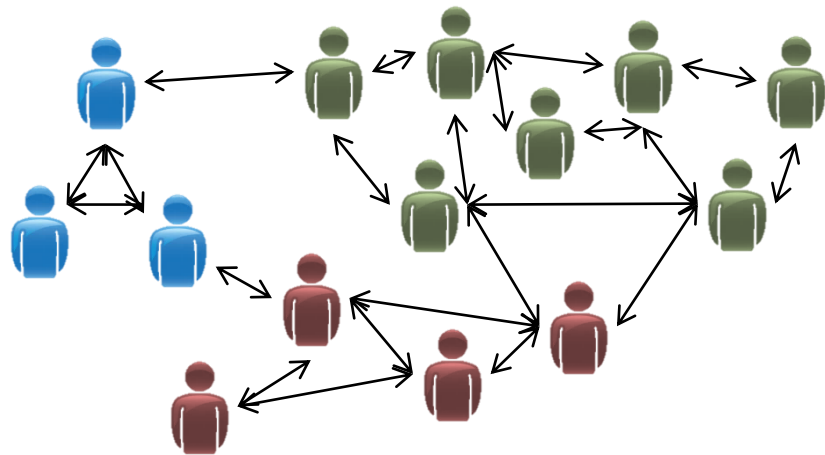


1. Britain keeps AAA credit rating

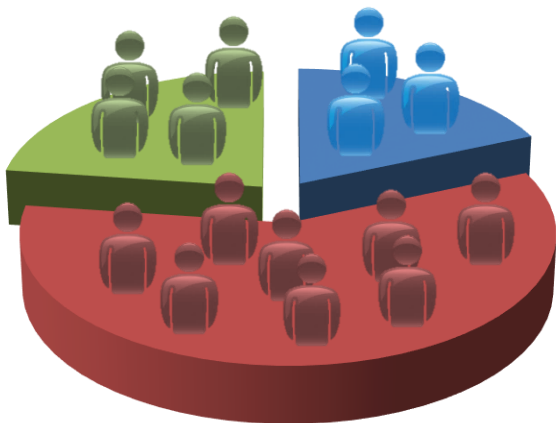




Organize computing clusters



Social network analysis

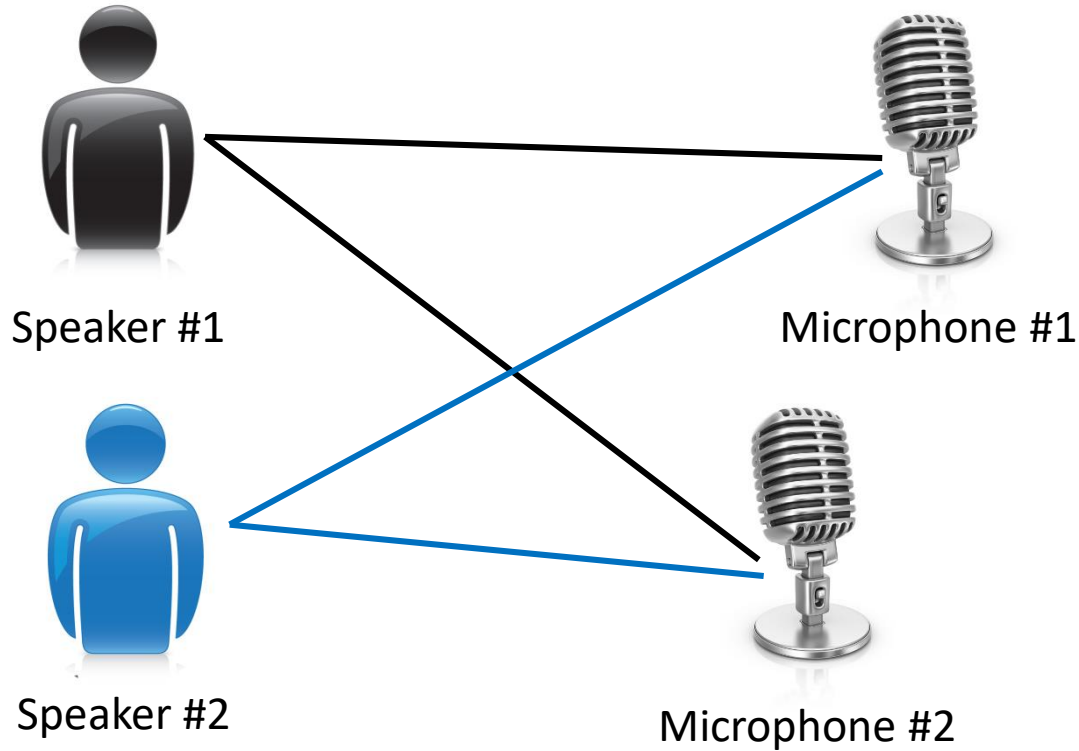


Market segmentation



Astronomical data analysis

Cocktail party problem



Cocktail party problem algorithm

```
[W,s,v] = svd(( repmat(sum(x.*x,1),size(x,1),1).*x)*x');
```


Of the following examples, which would you address using an unsupervised learning algorithm? (Check all that apply.)

Given email labeled as spam/not spam, learn a spam filter.

Given a set of news articles found on the web, group them into set of articles about the same story.

Given a database of customer data, automatically discover market segments and group customers into different market segments.

Given a dataset of patients diagnosed as either having diabetes or not, learn to classify new patients as having diabetes or not.

Course Contents – Machine Learning Techniques

Supervised Learning

- Linear Regression
- Logistic Regression
- Artificial Neural Network
- Support Vector Machine
- Naïve Baye's Classifier
- K-Nearest neighbor

Unsupervised Learning

- K-means Clustering, K-medoid Clustering
- Dimensionality Reduction - Principal Component Analysis (PCA)

Hypothesis Evaluation